Notes: Jovanovic and Rousseau (AER P&P, 2002)
The Q-Theory of Mergers

Contents

1 Big picture

- **Research question.** Can the same Q-theory that explains direct capital investment also explain mergers and acquisitions? More specifically, do high-Q firms acquire capital through M&A when the market value of that capital is low relative to the buyer's marginal value of capital?
- **Main idea.** Mergers are treated as transactions in a market for *used, bundled capital.* A firm can buy capital directly as new or disassembled used capital, or it can buy an entire firm and thereby acquire a bundle of capital through the M&A market.
- **Economic mechanism.** High-productivity or high-Q firms have a high marginal value of capital. Low-Q firms have capital that is more valuable inside a different firm. M&A reallocates capital from low-Q firms to high-Q firms.
- **Fixed-cost logic.** M&A has a fixed cost but a relatively low marginal adjustment cost. Small expansions are therefore done mostly through direct capital purchases, while large expansions rely increasingly on acquisitions.
- **Main empirical finding.** M&A investment responds more strongly to Q than direct investment does. In the baseline investment regressions, the coefficient on $Q - \bar{q}$ in the acquisition equation is nearly three times the coefficient on Q in the direct-investment equation.
- **Agency/free-cash-flow result.** Cash has little effect on direct investment but a positive and significant effect on acquisitions. This supports a limited Jensen-style free-cash-flow channel: managers with excess cash appear more likely to acquire, but Q still explains most of the action.
- **Merger-wave result.** Merger waves tend to occur when the dispersion of Q across firms is high. The waves of 1900, the 1920s, the 1980s, and the 1990s look like reallocation waves; the 1960s conglomerate wave is the main exception.
- **Economic takeaway.** The paper reframes mergers as a capital reallocation technology. M&A is not only a corporate-control event or agency problem; it is also a way for high-Q firms to obtain large amounts of capital quickly.

2 Incremental contribution of the paper

- **Extends Q-theory from internal investment to acquisitions.** Standard Q-theory says firms invest more when their marginal value of capital is high. This paper argues that the same logic applies to M&A investment.
- **Unifies two used-capital markets.** The paper links sales of disassembled used capital to M&A transactions. In both cases, existing capital is transferred to a new owner and, potentially, a better manager.
- **Explains why M&A is lumpy.** The model distinguishes direct investment x and acquisition investment y. Because acquisition has a fixed cost, the firm does not use y until the desired expansion is large enough.
- **Connects firm-level Q to aggregate merger waves.** The paper does not stop at cross-sectional investment regressions. It also asks whether historical merger waves line up with periods in which Q dispersion is high.

- **Separates reallocation and free-cash-flow explanations.** The authors show that cash matters for acquisitions, but Q remains much more important. This lets the paper acknowledge agency motives without making them the dominant explanation.
- **Short, model-driven empirical paper.** The paper is concise, but it combines theory, modern Compustat/CRSP evidence, and long historical merger-wave evidence.

3 Data and measurement

3.1 Capital investment measures

The paper uses firms common to CRSP and Compustat. The modern empirical analysis mainly covers 1971–2000.

- **Direct capital expenditures.** Direct investment is measured using Compustat capital expenditures, item 128.
- **Acquired capital.** Acquisition investment is measured using Compustat item 129, which records funds used for and costs related to acquisitions.
- **Capital sales.** Sales of disassembled used capital are measured with Compustat item 107, property, plant, and equipment sales.
- **Total investment.** For Figure 1, investment is the sum of acquired capital and direct capital expenditures.
- **Capital denominator.** Firm capital K is proxied by book assets, Compustat item 6, in the investment-rate regressions and in Figure 5.

Interpretation. The paper deliberately treats acquisitions as a form of investment. The central empirical comparison is not acquisition volume versus the number of deals, but acquisition spending versus direct capital expenditures.

3.2 Q and the price of acquired capital

The empirical proxy for firm Q is the market-to-book ratio.

- **Market value.** The paper constructs market values from the market value of common equity plus book preferred stock and debt.
- **Book value.** Book values use the book value of common equity plus book preferred stock and debt.
- **Acquired-capital price.** In the model, the acquisition-market price of capital is q. In the data, the authors use the average market-to-book value of disappearing firms as a proxy for $\bar{q}$.
- **Incentive to acquire.** The relevant acquisition incentive is $Q_{j,t-1} - \bar{q}_{t-1}$, not just $Q_{j,t-1}$, because the buyer compares its own value of capital with the price of capital in the acquisition market.

The paper notes an important measurement problem: target firms often write down book capital quickly, so observed target market-to-book values may overstate the true price of used capital in acquisitions. Because of this, Figure 5 and the regressions assume that one dollar spent on acquisitions buys the same efficiency units of capital as one dollar spent on direct investment.

3.3 Cash and agency measure

To test the free-cash-flow view, the paper adds cash normalized by firm capital:

$$100 \times \frac{\text{cash}_{j,t-1}}{K_{j,t-1}}. \tag{1}$$

Cash is Compustat item 1 and capital is again proxied by book assets.

Interpretation. If managers use excess cash for empire building, cash should predict acquisitions more than direct investment. Table 2 confirms this pattern.

3.4 Historical merger-wave data

The paper also studies longer-run merger waves.

- For the modern period, merger activity and firm Q measures come from CRSP and Compustat.
- For the historical series, acquisition activity uses CRSP after 1925 and data linked to Nelson's historical merger series for 1890–1930.
- Because Compustat does not provide reliable pre-1975 Q dispersion, the authors use a vintage-based proxy: for each vintage year t, they compute the dispersion of 1998 Q among firms founded in year t.

Interpretation. The vintage proxy is indirect. It relies on persistence in firm productivity and on the idea that cohorts born in high-dispersion periods retain information about those periods in their later Q distribution.

4 Model and empirical specifications

4.1 Production, capital accumulation, and costs

A firm has technology state z and capital stock K. Output is linear in capital:

$$\text{output} = zK. \tag{2}$$

The productivity state follows a firm-specific Markov process. The firm chooses direct investment X and acquired bundled capital Y. Next period's capital is

$$K' = (1-\delta)K + X + Y. \tag{3}$$

Define investment rates

$$x = \frac{X}{K}, \qquad y = \frac{Y}{K}. \tag{4}$$

The firm pays both the purchase cost of capital and a forgone-output adjustment cost

$$C(x, y)K. \tag{5}$$

Direct capital has price one. Acquired capital trades at a common price $q < 1$, equal to salvage value in the baseline model.

4.2 Firm value and Q

Because the production function and adjustment costs are homogeneous of degree one, firm value is linear in capital:

$$V(z, K) = Q(z)K. \tag{6}$$

The value per unit of capital satisfies

$$Q(z) = \max_{x\geq 0, y\geq 0} \left\{z - C(x, y) - x - qy + (1 - \delta + x + y)Q^*(z)\right\}, \tag{7}$$

where

$$Q^*(z) = \frac{1}{1+r} \int \max\{q, Q(z')\}\, dF(z', z). \tag{8}$$

The term $\max\{q, Q(z')\}$ captures the option to continue operating the capital inside the firm or sell it on the acquisition market.

At an interior optimum, the first-order conditions are

$$C_1(x, y) = Q^*(z) - 1, \tag{9}$$

$$C_2(x, y) = Q^*(z) - q. \tag{10}$$

The first condition governs direct investment. The second governs acquisition investment. Since $q < 1$, acquired capital can be attractive when the buyer's continuation value is high relative to the acquisition price.

4.3 Fixed cost of acquisitions

The paper adds a fixed cost ϕ of using the acquisitions market:

$$C(x, y) = \begin{cases} c(x, y) + \phi, & y > 0, \\ c(x, 0), & y = 0. \end{cases} \tag{11}$$

The threshold gross investment rate i^* solves

$$i + c(i, 0) = \phi + \min_{y} \left\{(i - y) + qy + c(i - y, y)\right\}. \tag{12}$$

For $i < i^*$, the fixed acquisition cost makes the firm use only direct investment. For $i > i^*$, the firm uses acquisitions as well.

Economic intuition. Direct investment is the natural margin for small expansions. Once the desired expansion is large enough, paying the fixed cost of acquisition becomes worthwhile, and the low marginal adjustment cost of M&A makes y rise rapidly.

4.4 Four firm regions

The model divides firms into four regions of z:

- **Exit/acquisition region.** If $Q(z) < q$, the capital is worth more outside the firm. The firm dissolves or is acquired.
- **Internal-only investment region.** For intermediate z, the firm remains active and sets $x > 0$, $y = 0$.
- **Internal plus external investment region.** For higher z, the firm sets both $x > 0$ and $y > 0$.
- **Acquisition-dominant region.** For the highest z, acquisition investment y exceeds direct investment x.

4.5 Investment regressions

The empirical investment equations linearize the model as

$$x_{j,t} = \frac{X_{j,t}}{K_{j,t-1}} = \alpha_0^x + \alpha_1^x Q_{j,t-1} + \alpha_2^x t + \text{industry effects} + u_{j,t}^x, \tag{13}$$

$$y_{j,t} = \frac{Y_{j,t}}{K_{j,t-1}} = \alpha_0^y + \alpha_1^y \left(Q_{j,t-1} - \bar{q}_{t-1}\right) + \alpha_2^y t + \text{industry effects} + u_{j,t}^y. \tag{14}$$

The model predicts $\alpha_1^x > 0$ and $\alpha_1^y > 0$. The paper multiplies the dependent variables by 100, so coefficients can be read as percentage-point responses.

The cash specification adds cash to both equations:

$$100x_{j,t} = \alpha_0^x + \alpha_1^x Q_{j,t-1} + \alpha_3^x 100 \times \text{cash}_{j,t-1}/K_{j,t-1} + \alpha_2^x t + u_{j,t}^x, \tag{15}$$

$$100y_{j,t} = \alpha_0^y + \alpha_1^y \left(Q_{j,t-1} - \bar{q}_{t-1}\right) + \alpha_3^y 100 \times \text{cash}_{j,t-1}/K_{j,t-1} + \alpha_2^y t + u_{j,t}^y. \tag{16}$$

5 Identification and interpretation

5.1 What the paper identifies

The paper identifies three empirical regularities that are consistent with the model:

- direct investment and acquisition investment both increase with Q-related incentives;
- acquisitions are more Q-sensitive than direct capital expenditures;
- merger waves line up with periods of high dispersion in Q.

The evidence is not a clean causal experiment. It is a theory-guided set of correlations. The identification strength comes from the model's sharp comparative statics, the contrast between x and y, and the historical alignment between Q dispersion and merger waves.

5.2 Why Q matters differently for x and y

In a standard Hayashi-style model, investment responds to Q because Q measures the marginal value of capital. Here the same logic applies to acquisitions, but y is expected to respond more strongly because acquisition has high fixed cost and low marginal adjustment cost.

Interpretation. A high-Q firm can grow incrementally with direct investment, but a large jump in capital is more naturally achieved through M&A. This is why the paper expects a larger coefficient in the y equation.

5.3 Agency interpretation of cash

The cash result is consistent with Jensen's free-cash-flow hypothesis: managers with excess cash may spend it on acquisitions even when acquisitions are not value maximizing. But the cash evidence does not overturn the reallocation story.

- Cash is insignificant or weak for direct investment.
- Cash is positive and significant for acquisitions.
- Q-related variables remain strongly significant after including cash.

Interpretation. Some mergers may be agency driven, but most of the systematic variation in acquisition spending is still related to Q-based reallocation incentives.

5.4 Interpreting merger waves

If all firms had the same z, then $Q = q$ and M&A would not reallocate capital. The model predicts that merger waves should occur when there is more dispersion in Q across firms, because the gains from moving capital from low-Q to high-Q firms are larger.

This logic leads to two tests:

- in the modern data, compare acquisition activity with the dispersion between high-acquisition firms' Q and the average Q of disappearing firms;
- in the historical data, compare acquisition waves with a vintage-based proxy for Q dispersion.

The modern correlation is modest contemporaneously but rises with lags. The historical vintage correlation is much larger, about 0.64 for the detrended series.

5.5 Main limitations

- **Q measurement.** Average Q may not equal marginal Q, and market-to-book ratios are noisy proxies for the model's $Q(z)$.
- **Target-book-value problem.** Acquired firms' book capital may be understated, causing observed target market-to-book values to overstate the true price q.
- **Acquisition heterogeneity.** Some M&A deals are reallocative, but others may be motivated by market power, diversification, tax, agency, or accounting motives.

- **Historical proxy.** The vintage-based Q-dispersion measure is indirect and may suffer from survivor bias because high-z firms are more likely to survive.
- **Short format.** As an AER Papers and Proceedings paper, the analysis is intentionally compact and does not provide a full structural estimation of the model.

6 Tables and figures

6.1 Figure 1 and Figure 2: Used capital and the four firm regions

Read Figure 1:

- The solid line plots acquired capital as a percentage of investment.
- The dashed line plots direct purchases of used capital as a percentage of investment.
- The two series move together with a correlation of 0.45.
- The share of investment spent on used capital rises from about 10 percent in 1975 to 43.5 percent in 2000.

Read Figure 2:

- Firms with very low z have $Q(z) < q$ and exit, dissolve, or are acquired.
- Firms with intermediate z invest internally only: $x > 0$, $y = 0$.
- High-z firms invest both internally and externally: $x > 0$, $y > 0$.
- The highest-z firms use acquisitions more than direct investment: $y > x > 0$.

Economic message: Figure 1 motivates treating M&A as part of the used-capital market. Figure 2 gives the model's central sorting result: low-Q capital exits or is acquired, while high-Q firms become buyers.

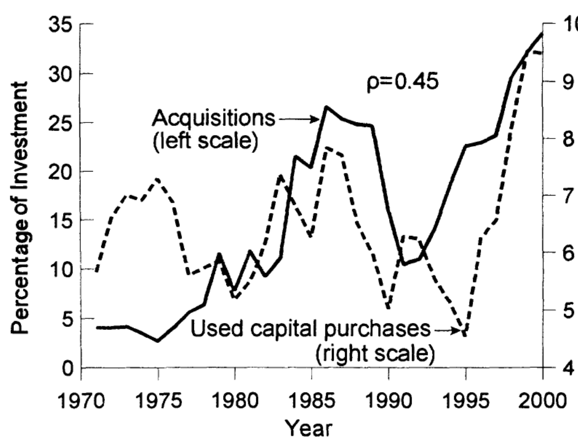

FIGURE 1. USED AND ACQUIRED CAPITAL AS PERCENTAGES OF TOTAL INVESTMENT, 1971–2000

Figure 1: Used and acquired capital as percentages of total investment.

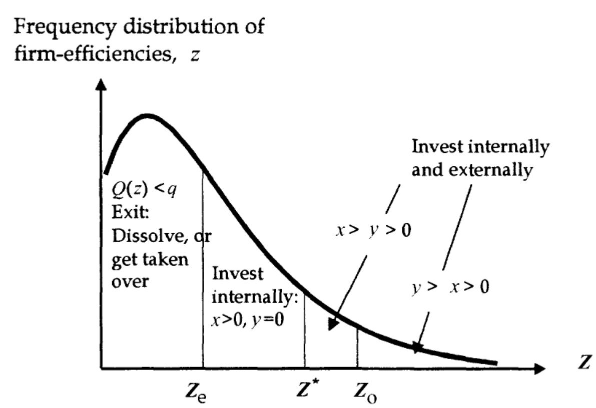

FIGURE 2. THE FOUR REGIONS OF z

Figure 2: The four regions of firm productivity z.

6.2 Figures 3 and 4: Expansion path and the overtaking point

Read Figure 3:

- The dashed iso-investment lines hold total desired investment $i = x + y$ fixed.
- For low i, the firm uses only direct investment.
- At the threshold i^*, acquisition investment jumps from zero to y_{min}.
- As i rises, the expansion path moves toward the 45-degree line because y becomes increasingly important.

Read Figure 4:

- Direct investment x initially rises with total investment i.
- At i^*, paying the fixed acquisition cost becomes worthwhile, so y jumps up and x drops.
- At i_o, acquisition investment overtakes direct investment.
- Beyond i_o, large expansions are mainly acquisition-based.

Economic message: The model predicts a nonlinear investment mix. Direct investment handles small expansions, but acquisitions dominate when the firm wants to expand by a large amount.

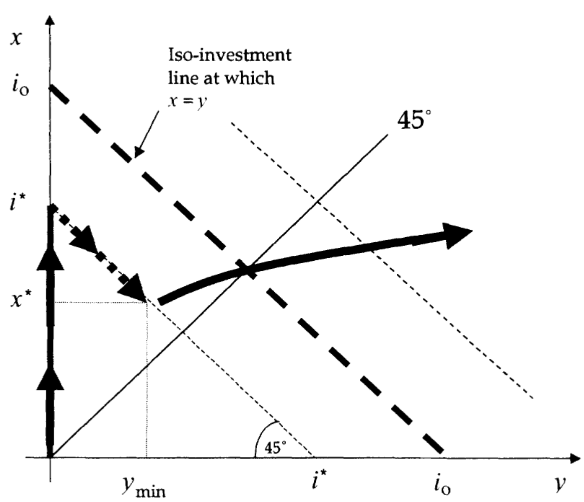

FIGURE 3. THE EXPANSION PATH OF x AND y AS THE GROSS INVESTMENT RATE, i, RISES

Figure 3: Expansion path of direct investment x and acquisition investment y.

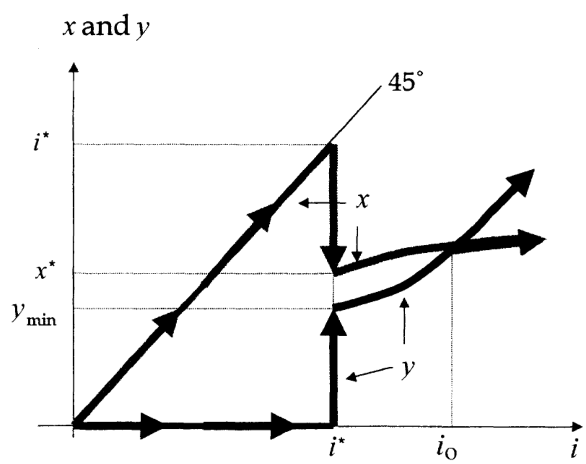

FIGURE 4. THE POINT OF OVERTAKING, i_o

Figure 4: Point where acquisitions overtake direct investment.

6.3 Figures 5 and 6: Empirical investment mix and Q groups

Read Figure 5:

- The figure plots HP-filtered means of x and y by total investment ratio $i = x + y$.

- Small expansions come mainly through direct capital purchases.
- Large expansions come mainly through acquisitions.
- Overtaking occurs at about $i_o = 1.12$, roughly a doubling of capacity after depreciation.
- The overtaking threshold falls over time: the paper reports 1.43 in 1980, 1.09 in 1989, and 0.5 in 1998.

Read Figure 6:

- The figure plots average Q for investment subgroups and the proxy for $\bar{q}$.
- All Q series are well above one, probably because target book capital understates true replacement cost.
- This motivates the paper's decision not to deflate acquisitions using the observed target market-to-book ratio.

Economic message: Figure 5 is the central micro-level visual support for the fixed-cost story. Figure 6 highlights the measurement issue in valuing acquired capital and explains why the paper treats dollars of x and y as comparable efficiency units.

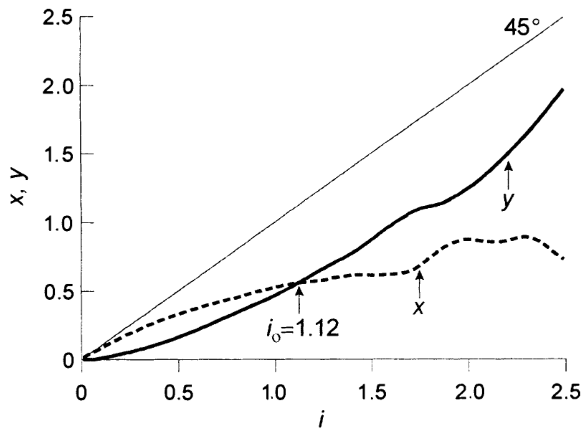

FIGURE 5. DIRECT CAPITAL PURCHASES (x) AND ACQUIRED CAPITAL (y), BY INVESTMENT RATIO, $i = x + y$, 1971–2000

Figure 5: Direct capital purchases and acquired capital by investment ratio.

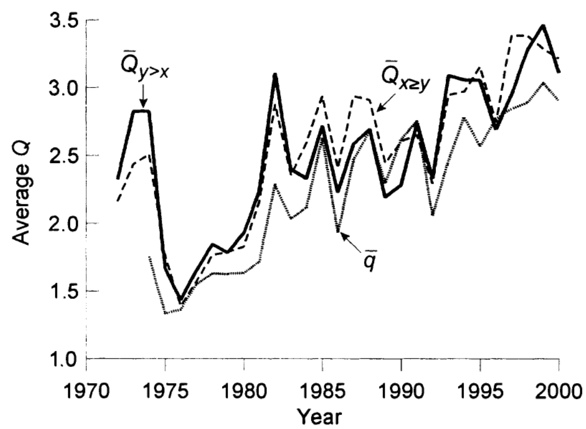

FIGURE 6. Q'S, BY INVESTMENT SUBGROUP, 1972–2000

Figure 6: Q measures by investment subgroup.

6.4 Tables 1 and 2: Investment regressions and cash

Read Table 1:

- The direct-investment coefficient on $Q_{j,t-1}$ is 0.746 with a t-statistic of 35.71.
- The acquisition coefficient on $Q_{j,t-1} - \bar{q}_{t-1}$ is 2.220 with a t-statistic of 18.42.
- The acquisition coefficient is nearly three times the direct-investment coefficient.
- The sample has 111,039 direct-investment observations and 26,383 acquisition observations.

Read Table 2:

- Adding cash barely changes the direct-investment Q coefficient: it remains 0.738.
- Cash has little effect on direct investment: coefficient 0.006, t-statistic 1.32.
- Cash has a positive and significant effect on acquisitions: coefficient 0.220, t-statistic 9.82.
- The acquisition Q coefficient remains economically large and statistically significant: 1.916, t-statistic 15.43.

Economic message: Table 1 supports the Q-theory of mergers. Table 2 shows that free cash flow helps explain some acquisition spending, but Q-based reallocation remains the dominant empirical force.

TABLE 1—INVESTMENT REGRESSIONS

	Dependent variable	
Independent variable	$100x_{j,t}$	$100y_{j,t}$
$Q_{j,t-1}$	0.746	
	(35.71)	
$Q_{j,t-1} - \bar{q}_{t-1}$		2.220
		(18.42)
Time trend	−0.120	0.0308
	(13.29)	(7.32)
R^2:	0.0479	0.0206
N:	111,039	26,383

Notes: The table presents estimates for equation (10) with t statistics in parentheses. The regressions include dummy variables for two-digit SIC's (not reported).

Table 1: Investment regressions.

TABLE 2—INVESTMENT REGRESSIONS WITH CASH

	Dependent variable	
Independent variable	$100x_{j,t}$	$100y_{j,t}$
$Q_{j,t-1}$	0.738	
	(34.41)	
$Q_{j,t-1} - \bar{q}_{t-1}$		1.916
		(15.43)
$100 \times \text{cash}_{j,t-1}$	0.006	0.220
	(1.32)	(9.82)
Time trend	−0.121	0.277
	(13.38)	(6.59)
R^2:	0.0479	0.0241
N:	111,039	26,383

Notes: The table presents estimates for equation (10), with the ratio of cash to total firm assets as an additional regressor, with t statistics in parentheses. The regressions include dummy variables for two-digit SIC's (not reported).

Table 2: Investment regressions with cash.

6.5 Figures 7 and 8: Q dispersion and merger waves

Read Figure 7:

- The figure compares acquisition activity with a detrended measure of Q dispersion, $\bar{Q}_{y>x} - \bar{q}$.
- The contemporaneous/one-year-ahead timing shown has a correlation of 0.12.
- The correlation rises to 0.22 with another lag and 0.31 with one more lag.
- The message is that high Q dispersion tends to precede more acquisitions.

Read Figure 8:

- The solid line plots historical acquired capital as a percentage of total stock-market capitalization.
- The dashed line plots vintage-based dispersion of Q.
- The detrended correlation between the two series is 0.64.

- The 1900, 1920s, 1980s, and 1990s waves line up with high Q dispersion; the 1960s wave is the main exception.

Economic message: Merger waves are not all alike. Many historical waves look like periods when capital reallocation opportunities are unusually large. The 1960s conglomerate wave is less consistent with the Q-theory reallocation story.

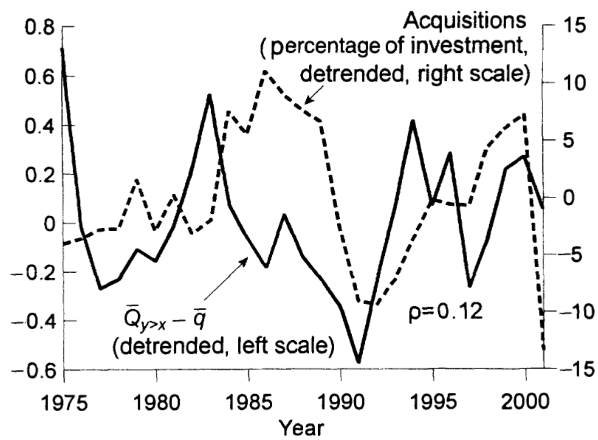

FIGURE 7. ACQUIRED CAPITAL AND THE DISPERSION OF Q, 1975–2001

Figure 7: Acquired capital and Q dispersion, 1975–2001.

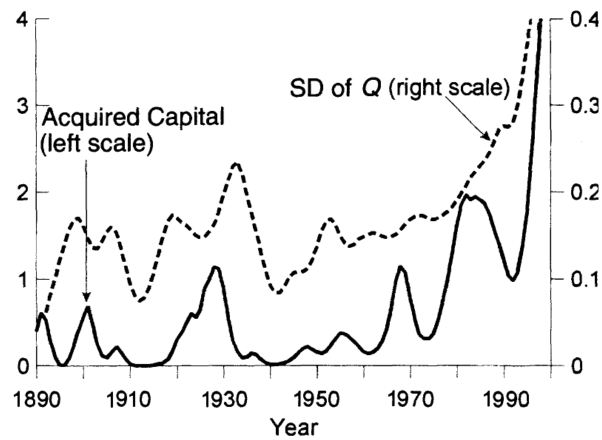

FIGURE 8. ACQUIRED CAPITAL (AS A PERCENTAGE OF TOTAL STOCK-MARKET CAPITALIZATION) AND THE DISPERSION OF Q (BY VINTAGE), 1890–1998

Figure 8: Historical acquired capital and vintage Q dispersion, 1890–1998.

7 Short summary

Jovanovic and Rousseau extend Q-theory to mergers. A high-Q firm has a high value of additional capital. It can expand by purchasing new or disassembled capital, or by acquiring an entire firm and thereby obtaining bundled used capital. Since acquisitions have a fixed cost but relatively low marginal adjustment cost, the model predicts that small expansions use direct investment while large expansions use acquisitions.

The empirical evidence fits this logic. Acquisition investment is more responsive to Q than direct investment is. The coefficient on the acquisition incentive $Q - \bar{q}$ is nearly three times the coefficient on Q in the direct-investment equation. Cash predicts acquisitions but not direct investment, consistent with some free-cash-flow behavior. However, Q remains strongly significant after controlling for cash.

At the aggregate level, merger waves tend to occur when Q dispersion is high. This supports the view that many merger waves are reallocation waves: capital moves from low-Q firms to high-Q firms. The main exception is the 1960s wave, which the paper interprets as likely driven by something other than profitable reallocation opportunities.

8 Conclusion

8.1 Core conclusion

The paper's core conclusion is that M&A can be understood as an investment technology. High-Q firms acquire low-Q firms because acquisitions move capital to better uses. The same Q-theory that explains direct investment also helps explain acquisition investment.

8.2 Economic intuition

When a firm's productivity state z is high, its capital value $Q(z)$ is high. Such a firm wants to grow. If it wants to grow a little, it buys capital directly. If it wants to grow a lot, it pays the fixed cost of acquisition and buys bundled capital through M&A. Low-productivity firms, whose capital is worth more outside the firm, exit or become targets.

8.3 Incremental contribution revisited

The contribution is not simply saying that high-Q firms buy low-Q firms. The paper embeds that fact in a model where acquisition is a margin of investment with fixed and marginal adjustment costs, and then links the model to firm-level investment regressions and long historical merger waves.

8.4 Implications

- **For corporate finance.** Mergers should not always be interpreted as agency events. Many are part of the real allocation of capital.
- **For investment theory.** Q-theory applies to both direct investment and acquisition investment, but the two margins have different adjustment-cost structures.
- **For merger waves.** Merger waves can reflect periods of unusually high reallocative gains, not just market timing, hubris, or empire building.
- **For empirical work.** Acquisition spending should be treated as part of a firm's investment policy, especially when studying large expansions.

8.5 Limitations

The paper is best read as a compact theory-and-evidence piece rather than a definitive causal analysis. It relies on market-to-book measures of Q, noisy acquisition data, and indirect historical proxies for Q dispersion. It also abstracts from many other motives for M&A, including market power, diversification, taxes, managerial hubris, and financing constraints.

8.6 Takeaway

Mergers reallocate capital. Acquisition investment is more sensitive to Q than direct investment, and merger waves tend to arrive when Q dispersion is high.